The Causal Approach to Algorithmic (Un)Fairness

Can a causal approach overcome the limitations of conventional statistical approaches to improving the fairness of algorithms supporting decision-making?

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ABSTRACT

Machine learning algorithms are increasingly used to inform important decisions, including hiring, giving bail, and granting loans. Recently, scandals of blatantly unfair algorithms have caught the public eye and spurred data scientists to develop methods that ensure algorithmic fairness. While multiple statistical methods and definitions of algorithmic fairness have appeared, they are not all mutually compatible and there is currently no consensus regarding which methods or definitions truly guarantee fairness. We argue here that these statistical approaches are sometimes insufficient because they fail to consider the causes of discrimination within data. They make implicit assumptions about the causal structure of the data, and about what constitutes a fair causal relationship. We therefore investigate an alternative, causal approach to algorithmic fairness that relies on causal models. Causal approaches to fairness focus on ensuring that algorithms do not rely on data which reflects unfair causal relationships. We present Counterfactual Fairness, the first causal definition of fairness, to illustrate the principle advantages and limitations of the causal approach. According to Counterfactual Fairness, an algorithmic prediction for an individual is fair if it would have been the same if the individual's designated 'protected feature' (e.g. sometimes 'race' or 'gender') was different. Though we show that the limitations of statistical approaches and the nature of discriminatory processes suggest a causal approach, this does not justify the normative aspect of the causal approach to fairness. We therefore also discuss whether philosophical theories of justice and human's fairness perception support this approach.

PREFACE

This dissertation discusses the problem of algorithmic fairness and the value of a new, causal approach.

Algorithms can be used to predict how people will act and think. These predictions can then inform decisions, such as hiring a job candidate or granting loans. Because of the impact of such decisions, it is important that these algorithms be perceived as fair by the public. Yet in several recent publicised cases, algorithm-aided decisions have been criticised for showing gender or racial bias.

Data scientists have proposed many methods to improve algorithmic fairness. But their progress is limited by the difficulty of agreeing on a statistical definition of what constitutes a 'fair' prediction. This results from a lack of consideration for the different possible causes of discrimination. I therefore investigate a recent causal approach to algorithmic fairness, which requires data-scientists and decision-makers to explicitly consider the causal factors contributing to a situation, then to evaluate whether their influence on the algorithm's prediction is fair.

I was drawn to subjects concerning morality and fairness following my internship in a computational psychology lab at McGill University. During this internship, my supervisor and I showed that behaviours observed in the ultimatum game, widely considered as evidence for humans' preference for fairness, could arise from purely self-interested decisions, given my supervisor' decision-making model. That summer, I also took an online course on machine learning, because its ubiquity makes its apprehension increasingly necessary to understand new technologies and decisions, and because it is a valuable tool for modelling and statistical analysis in my fields of interest: neuroscience and cognitive sciences. Algorithmic fairness thus united two recent interests of mine, and enabled me to investigate a topic of growing concern in society – as evidenced by the exponential amount of attention it has received in academia and in the media. Finally, my dialogue with my supervisor, Professor David Lagnado from UCL's Causal Cognition lab, directed my search to existing causal approaches to the subject.

I consider that papers presenting new methods for algorithmic fairness often do not provide sufficient justification to legitimise their approaches. Therefore, before discussing the causal approach, I explore whether it is supported by people's perception of fairness and philosophical

theories of justice. I refer to philosophical views because data scientists and decision-makers involved have generally not thought about these subjects in as much depth and with as large a scope as philosophers; they might be biased by their personal experience and the interests of the company/institution they represent. Framing a new approach to algorithmic fairness within its philosophical context both provides supporting arguments and highlights the trade-offs it implies.

To discuss algorithmic (un)fairness we therefore draw on statistics – required for implementing fairness in machine learning algorithms, political philosophy – to discuss established concepts of justice, their implications and the importance of causality, moral psychology – to consider laymen's perception of fairness, and the recently mathematically formalised domain of 'causal inference' – for the framework it provides to conduct causal and counterfactual analysis.

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1 INTRODUCTION

This dissertation considers the problem of ensuring fairness in machine learning(ML)*[1] algorithms*. The inherent bias in data, limitations of ML and difficulty of defining fairness account for the persistence of algorithmic unfairness. Adopting a causal approach, which focuses on cause-effect relationships underlying the data, allows us to overcome the field's recent deadlock.

What are ML algorithms?

Algorithms are sequences of steps generally performed by a computer. ML algorithms (from now on 'algorithms' refer to 'ML algorithms') can identify patterns in data with only limited human assistance (1). They therefore compensate for humans' inability to make predictions from large data sets and bypass the need to call on experts for individual decisions (1). Instead of being given detailed, explicit instructions on how to treat the data, algorithms are given sparse guidelines to direct how they learn from the data and to increase their predictions' accuracy (1). These guidelines are applied to training data* containing the characteristics, called 'features'*, of a sample of the population. Algorithms thus create a statistical model of how features map onto the predicted outcomes. For example, if the goal is predicting a prisoner's risk of recidivism to inform bail decisions, features may include a convict's age, number of convictions, and the severity of their crimes; the training set would be composed of numerical values representing these features for a large number of convicts previously considered for bail (1). Algorithms, when combined with vast amounts of data and computational power, are powerful tools for predicting an individual's future behaviour, such as the probability that a prisoner will reoffend if given bail, and for directly classifying individuals into different groups, for example distinguishing university applicants who are 'likely' or 'unlikely' to successfully complete a degree (1).

[1] The definition of words followed by a (*) can be found in the Appendix, Section 7.1.

Why do we entrust decisions to algorithms?

Humans have a limited memory (2,3), attention (4–7) and information processing capacity (6,8), so computationally complex evaluative tasks such as those mentioned above are both difficult and time-consuming. Therefore, relying on algorithms' predictions to make decisions is tempting.

Human decision-making is also subject to many biases, such as the anchoring effect* (9), availability heuristic* (10,11), confirmation bias* (12) and halo effect* (13,14). For example, in the medical field, doctors have been shown to neglect base rates* leading to misinterpreted test results (15) and to be influenced by the framing* of information (16), and judges' decisions can be inconsistent (17) and influenced by their race[2] (18) and religion (19,20). Belle, Cantarelli & Belardinelli (21) also show a systematic influence of anchoring and halo effects on performance reviews of employees by public sector managers.

By contrast, algorithms seem to offer consistent and unbiased judgement. Because they seemingly provide a more reliable alternative, algorithms are increasingly used to inform decision-making during qualitative assessments, for example to give bail (22), to grant loans (23), during hiring decisions (24,25) and perhaps during academic admissions (26).

Are we right to entrust decisions to algorithms?

The four main concerns associated with the increased use of algorithms include: data protection, job replacement, transparency and fairness. While the first two have been widely canvased for more than a decade, transparency and fairness only gained traction in the last years (1). Interest in fairness was catalysed by publicised cases of supposedly biased algorithms including the controversy around the Correctional Offender Management Profiling for Alternative Sanctions (COMPAS) algorithm (1,27). COMPAS provides an estimate of a prisoner's risk of recidivism and is used by several U.S courts to inform bail decisions (22). In 2016, Pro-Publica, an investigative journalism company accused COMPAS of racial bias because it mistakenly labelled more black prisoners than white as having a high recidivism likelihood, and mistakenly

[2] Throughout this work, 'race' refers to the widespread perception of different races, which affects the treatment of people thus characterized. See the definition of 'race' in the Appendix, Section 7.1, for a more in depth characterisation.

labelled more white prisoners than black as having a low recidivism likelihood (1,22). This case drew the attention of the national press (28) and academic circles, who debated whether COMPAS showed bias or whether this was an inevitable artefact of unequal base rates* (more black prisoners than white). Other cases include gender bias in Amazon's hiring algorithm (25) and in Google's translation software (29), and racial bias in algorithms assessing 'creditworthiness' to inform loan allocations (23) and in Optum's (a healthcare provider) algorithm assessing health requirements (30). Throughout this work we rely on examples of algorithms which inform bail, admissions and hiring decisions to illustrate fairness definitions and limitations of ML.

These controversies have highlighted the need for strategies to assess and implement algorithmic fairness and therefore to agree on a definition of a 'fair prediction' (31).

Since 2017, a few research groups (32–37) have started adopting a new, 'causal' approach to defining and implementing algorithmic fairness, which takes into account the complex causal relationships which underlie the data provided to an algorithm.

Outline:

To understand why this approach is a valuable addition to the field of algorithmic fairness we must first consider: what is the current state of the field? Does a causal approach yield equally controversial definitions or does it address the fundamental problems which cause the current deadlock in the field?

We start with an overview of the steps within the algorithmic process which can be targeted to improve fairness. This discussion presents the principle causes of algorithmic biases, which emphasizes the complexity of algorithmic unfairness. Studying the principle statistical definitions of fairness and their important limitations then reveals the main challenge for algorithmic fairness: the need to clarify and agree on assumptions about the data in each scenario before debating the fairest procedure. We explain why the causal approach is best suited to address this problem which is closely tied to the shortcomings of conventional statistics.

Because most statistical papers on algorithmic fairness have consigned notions of what a 'fair' causal relationship is to implicit assumptions, they neglect two important sources of insight on the subject. The first is political philosophy, which accumulates 2000 years of reflections on

Justice, and has been particularly prolific in the last 50 years (revived by Rawls' 1971 'A Theory of Justice' (38)). The second is humans' perception of fairness, which matters because algorithms have increasingly large impacts on people's lives and because the perception of algorithmic unfairness can translate to a mistrust of the institutions using them (e.g. a state's judicial system). We therefore consider a few important views and findings in both domains, and whether they support a causal approach to fairness.

We introduce the recent mathematical formalisation of causal relationships and causal inference, on which rely causal approaches to algorithmic fairness. These approaches focus on ensuring that algorithms do not rely on data which reflects unfair causal relationships. To illustrate this, we present the first and most discussed causal definition of algorithmic fairness: Counterfactual Fairness. Counterfactual Fairness puts forward a quantitative method for evaluating the fairness of specific cause-effect relationships between the features and the prediction. This definition is developed through examples and a discussion of its advantages and limitations compared to previous statistical definitions.

2 METHODOLOGY

Algorithmic fairness necessarily relies on data science, requiring knowledge of ML to integrate notions of fairness into the algorithms. But it also requires insights from political philosophy – which provides bases for normative approaches to fairness, from psychology – to consider the public's perception of fairness, from the judicial field – because there is insufficient legal support to prevent common forms of algorithmic discrimination[3], and finally, as we will argue, from a mathematical formalisation of causal inference. Below we present the type of literature relied upon in each section and the justification for this selection.

Section	Resources/Search Criteria	Justification
3.1 **Source of algorithmic bias**	- Key-word driven search on Google Scholar and UCL's Online Library using combinations of the following: 'AI', 'machine learning', 'ML', 'big data', 'algorithmic', 'classification', 'prediction', 'implementing', 'fairness', 'fair', 'bias' and 'discrimination' - Papers from Conferences on Fairness, Accountability, and Transparency (FaccT) (41) - Focus on reviews of the field (40,42–45) and original papers as needed to clarify specific topics	- Algorithmic fairness is a prolific field so criteria is required to narrow down the literature search. - A wide collection of readings is needed to identify all major reviews and have resources for topic-specific inquiries.
3.2.1-3.2.3 **Statistical definitions of fairness**	- Review articles on statistical definitions of fairness (46,47) - Original papers of authors presenting new statistical definitions of fairness Note: there is no unified terminology to designate definitions so we cite papers using the same terminology as us	- Reviews present classifications of definitions and use consistent terminology and notation to present definitions (unlike the many original papers) which simplifies their comparison - Original papers discuss limitations of other definitions to show that their definition overcomes them
3.2.4 **Worldviews**	- First paper to discuss worldviews (48)	- (49,50) reinterpret worldviews to overcome certain limitations of (48) but we exclude them for simplicity and because they create new limitations
4.1 **Philosophical approaches to fairness**	- Focus on Rawls' Contractarianism, Scanlon's Contractualism, Mill's Utilitarianism and Dworkin's Egalitarianism - Original texts - Stanford's Encyclopaedia of Philosophy (SEP), whose entries are written by academics specialised in relevant philosophical domains, and reviewed by an editorial board (see (51)).	- Political philosophy is a very wide field; to narrow our discussion we chose four seminal theories of justice - We rely on SEP and original texts to familiarise ourselves with these views - Political philosophy papers discussing algorithmic fairness are rare (52–54), (52) was particularly

[3] A legal perspective on algorithmic fairness is beyond the scope of this work; see (39,40) for compelling introductions to the subject.

	- Political philosophy papers explicitly discussing algorithmic fairness Note: philosophical views discussed are complex and have many interpretations; we only seek to engage a discussion on the matter, drawing out their main principles and assessing the importance given to causal relationships.	relevant to our consideration of causal concerns in Egalitarianism
4.2 **Psychological approaches to fairness**	Fairness in children: - draw general conclusions supported by at least two psychological studies Fairness in adults: - meta-analyses of behaviour in the ultimatum game (55) and dictator game (a variant of the previous game) (56) - individual studies which come to the same conclusion (57–59) - one study explicitly considering the role of causal relationships in fairness perception (60) - two studies explicitly considering perception of algorithmic fairness (31,61)	Fairness in children: - Why? Young children's perception reveals more about instinctive components of fairness perception - methods examining children's fairness perception vary greatly so meta-analyses not available Fairness in adults - Why? Adults may be closely impacted by algorithmic decision-making - a canonical task for evaluating fairness in adults is the ultimatum game, so we can rely on meta-analyses of ultimatum game behaviour - we identify only one study linking causal relationships and fairness perception - we identify four studies on algorithmic fairness perception but exclude two (62,63), which, when comparing preferences for different fairness definitions, give a narrow (and non-overlapping) choice of definitions, so we cannot draw meaningful conclusions
5.1 **Causal inference in statistics**	- Judea Pearl's scientific publications (64,65) and latest book (66)	- Judea Pearl developed the most common framework for formalising causal inference in statistics
5.2 **Counterfactual Fairness**	- Original paper on Counterfactual Fairness (CF) (32)	- CF is the first fairness definition relying on a causal approach and also the most discussed
5.3 **Critical assessment of Counterfactual Fairness**	- Papers discussing CF and possible alternatives (36,37,67–69)	- There are few applications of the causal approach to algorithmic fairness so we surveyed the entire literature for critical discussions of CF (including the authors' revision of CF (67)) and to present alternative causal approaches

Figure 1: Research Methodology

3 STATISTICAL APPROACHES TO ALGORITHMIC FAIRNESS

Section 3 provides an overview of statistical approaches to algorithmic fairness in order to contextualise and justify the emergence of the causal approach. Section 3.1 introduces the different stages of the algorithmic process targeted by fairness implementation methods: 'pre-processing', which concerns constructing the training data provided to the model, 'in-processing', which modifies how the algorithm learns from data, and 'post-processing' which changes the algorithm's output (see *Figure 1*). We also examine principal sources of algorithmic bias, which all arise in the pre-processing stage. Section 3.2 presents statistical fairness definitions and highlights their limitations. Finally, in Section 3.3 we explore some assumptions underlying these definitions, and untangle the reasons for the lack of consensus regarding statistical fairness definitions.

Throughout the paper we use the following terminology:

Target variable*: what the algorithm tries to predict to inform a certain decision. 'Likelihood of recidivism' and 'likelihood of passing first year exams' can be target variables to inform bail decisions and college admissions respectively.

Protected attributes (or features)*: features whose influence on a decision or outcome is considered unfair. A **protected class*** is any group defined by the possession of a protected attribute.

Features*: non-protected features in the data used to predict the target variable, for example 'number of arrests' can be used to predict recidivism.

Characteristics of interests*: features a decision-maker is interested in but that are difficult to measure, so evaluated indirectly through more accessible features, for example 'math knowledge' might be evaluated through 'math A-level score'.

Outcome*: the reality behind the target variable. During training, algorithms generally compare their prediction of the target variable to the actual outcome found in the training data.

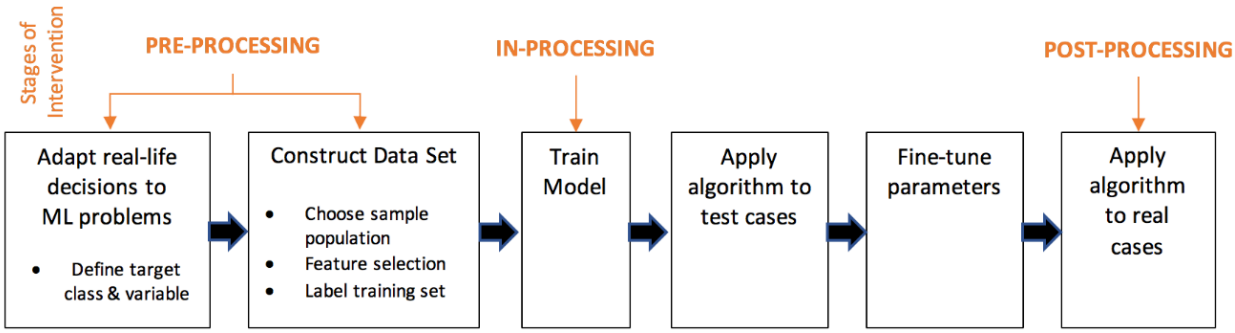

Figure 2: Flowchart of the algorithmic process.

Note: Bullet points indicate tasks in which bias can arise. Orange lettering identifies the stages of interventions.

3.1 Implementing algorithmic fairness

3.1.1 Pre-processing

An algorithm uses training data to learn its own mathematical model of how features in the data correlate with the outcome. This involves a process of induction: drawing rules from examples and generalising them to new cases. Therefore, an algorithm is only as good as its training data (40). Problematically, historical examples that algorithms train on often reflect historical prejudices and demographic inequalities. Training data being the principal source of algorithmic unfairness, we examine different ways in which it can introduce bias.

Defining the target variables and class labels: subjectivity in problem specification

Algorithms are often requested by non-data-scientists to aid decision-making. Data scientists must translate a real-world problem into a ML problem. This requires choosing a target variable and thresholds to classify numerical outputs meaningfully (40). Target variables must be relevant to the decisions informed by the algorithm, and thresholds must categorise the numerical probabilities output by the algorithm into relevant classes, for example dividing probabilities of recidivism into a high, medium and low recidivism risk. However, this is a subjective process, and data scientists unfamiliar with the field of application can unintentionally formalise the problem such that a protected class is (systematically) disadvantaged (40). For example, using 'estimated tenure' as a target variable to inform hiring

decisions has a greater disparate impact on protected classes than relying on predicted work productivity (40).

Labelling examples

Most prediction algorithms rely on supervised learning*. Their training data contains examples, each with a set of features associated to a past outcome (a 'label'). An algorithms predicts the probability that a certain outcome is associated to these features, and compares its prediction to the actual outcome. For example, if it predicts a prisoner has a 0.90 recidivism risk, it will check whether this prisoner did reoffend. But when the actual outcome is drawn from the past, there is a high risk of inheriting prior prejudice. For example, training an algorithm on past decisions regarding student admissions led St. George's Hospital's medical school to systematically discriminate against female racial minority applicants (with the same qualifications as other applicants) (70). Therefore, while outcomes in the data are used as ground truth, they can easily reflect conscious or unconscious biases of past decision-makers.

Data Collection

Algorithms also display bias when the representativeness of examples it trains on co-varies with the protected class, for example if a protected class is underrepresented or overrepresented. Data scientists use large databases to increase the accuracy of the inferential process. If protected classes are underrepresented, algorithms will make less accurate generalisations about feature-outcome associations, and therefore make biased predictions for this protected class. When predicting recidivism, because racial minority prisoners in past data are less likely to receive bail due to racial prejudice, there is less information on whether they reoffend when given bail, so generalisations are less robust for racial minority prisoners. The 'St. George's Hospital's medical school admissions' example in the previous section is also relevant here. Not only did past decisions about female and racial minority applicants reflect past discrimination, there were probably fewer past examples of applicants from these two classes for the algorithm to train on, making predictions less accurate for future candidates from these classes. Overrepresentation also cause unfair outcomes. If police men or managers pay closer attention to a certain protected class, they inevitably observe more offences or mistakes by its members than if they observed all classes equally. Relying on this information to direct their attention in the future then starts a self-justifying, positive feedback loop: increased attention

causes them to notice more mistakes which justifies focusing their attention on this protected class etc.

Feature Selection

Algorithms predict outcomes by extracting correlations between features and outcomes, but the features it relies on are reductive representations of complex processes. The coarseness of the features can lead to less accurate predictions for a protected classes. Details needed to reach equal accuracy are sometimes hard to obtain and come at an increased cost. For example, passing certain standardized tests heavily relies on test-specific practice. Therefore, students whose schools did not organize extensive practice (e.g. less wealthy schools) will be systematically disadvantaged if a university relies on such grades to assess their subject knowledge.

This highlights a problem: protected groups often possess different distributions of seemingly useful features (e.g. criminality rates, standardized test scores). While it may be statistically sound to rely on the protected attribute for predictions, it is still inaccurate because the existence of a correlation does not justify the generalisation of a relationship (71); it is possible that several external factors mediate a feature-outcome correlation, indicating a need for more granular data.

Such problems have spurred the development of pre-processing methods (see (44,72) for an overview) to relabel examples fairly (73,74), re-weight* certain features (75) and modify training samples to increase representativeness (76).

3.1.2 In-processing

Fairness implementation methods can target the algorithmic learning process, although bias does not generally arise there. Algorithms have an accuracy constraint: when predicting training examples, algorithms try to minimise a function representing their prediction errors. Most in-processing techniques involve adding an additional mathematical constraint which requires the function to minimise the unfairness of their result at the same time as the inaccuracy (see (77–80) for examples).

3.1.3 Post-processing

Finally, outcomes of existing algorithms can be modified to enforce a statistical definition of fairness such as Equalized Odds (81) and Calibration (82) (definitions described in Section 3.2.1). They can easily be applied to different algorithms but are sometimes insufficient because they do not consider the causes of the problem.

While pre-processing methods treat possible sources of the problem, they do not address all sources, thereby allowing persistent disparate impact. In- and post-processing methods are both dependent on the validity of their underlying definitions of fairness, which, as we will see next, is not straightforward.

3.2 Statistical definitions of fairness

Finding an optimal statistical measure of fairness is necessary to assess the fairness of algorithms and to guide fairness implementation methods. Although many publications offering their own definitions have emerged in the past three years, there is currently no consensus regarding which definition best captures fairness.

There are different ways of grouping existing definitions, but a simple, meaningful distinction is between group-level and individual-level definitions of fairness. Group-level definitions compare certain statistical metrics across protected groups. Individual-level definitions require similar individuals to receive similar predictions. Below we present examples of statistical fairness definitions and their limitations (see Appendix, Section 7.2.2 for these definitions' mathematical formulation).

3.2.1 Group Fairness

Demographic Parity requires the protected attribute and the algorithm's prediction to be statistically independent, implying that people from either protected class should have the same probability of receiving a positive* prediction (81). In the case of COMPAS, the probability of being considered 'likely to reoffend' (here the 'positive' outcome) should therefore be the same for white and non-white prisoners (the protected classes). However, picking the prisoners most likely to reoffend in one group and picking the same proportion at random in the other group would satisfy Demographic Parity. Demographic Parity can be satisfied in evidently unfair

ways, therefore it does not guarantee fairness. In addition, it cripples ML because it prevents the use of relevant features that happen to be possessed in different proportions in both groups. For example it proscribes using the feature 'crime linked to drug addiction' to predict recidivism if its base rate differs across groups. In response, Corbett-Davies et al (83) created **Conditional Statistical Parity** which requires different groups to have the same likelihood of receiving a positive prediction, conditional on a set of legitimate features.

Other definitions impose similar accuracy and inaccuracy rates across groups. They use error measures such as false positive, false negative, true positive and true negative rates (see Appendix, Section 7.2.1 for their definitions). **Predictive Parity,** for example, requires an equal proportion of correctly predicted positive outcomes in both groups (47,84), suggesting that people who receive a positive prediction should show the same real outcomes across groups. For example, if an algorithm predicts students' 'likelihood of passing first year exams' to inform admission decisions, the proportion of students considered as 'likely to pass first year exams' who did pass first year exams should be the same for all protected classes.[4] Two similar definitions (**Equal Opportunity** and **Equalized Odds** (81)) consider that people with the same real outcome should be treated in the same way by the algorithm (receive the same prediction), regardless of their group.

A final example of a group-level fairness definition is **Calibration** (84), which stipulates that for a given numerical prediction made by the algorithm, both groups should have the same probability of receiving a certain label. For example, if 90% of white prisoners with a 0.3 likelihood of recidivism are considered 'low recidivism risk', 90% of black prisoners with a 0.3 score should receive this label too.

3.2.2 Individual Fairness

Individual definitions of fairness impose a similar treatment of individuals according to some criteria.

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[4] We do not use the COMPAS example here because prisoners found 'likely to reoffend' do not usually receive bail and therefore cannot falsify this prediction.

Fairness Through Unawareness (32,85) requires algorithms not to use protected attributes to make predictions. While this is intuitive, many other features can be highly correlated with protected attributes (40), so removing them (as is sometimes required by law) does not guard against their influence on the prediction.

A seminal definition of algorithmic fairness is **Fairness through Awareness** (86) according to which similar individuals should receive similar predictions. This involves comparing the similarity between individuals' features and imposing a corresponding minimal similarity for the algorithm's predictions for these individuals. However, establishing a similarity metric for features has proven difficult and controversial. Indeed, prisoners and job applicants may display completely different sets of features making it difficult to establish equivalences.

3.2.3 Limitations of statistical measures of fairness

Many of the fairness definitions above cannot be satisfied simultaneously except in exceptional circumstances. For example, it is impossible to simultaneously satisfy Equalized Odds and Calibration (87), or two definitions relying on different inaccuracy measures (such as false positive rate, false discovery rate etc.) (82), or Predictive Parity, Equal Opportunity and Equalized Odds when the base rates differ across protected groups (88). Therefore, definitions must be assessed critically to find the most robust.

This brings us to another trade-off. An algorithm originally only tries to maximize its accuracy during the learning process, so imposing a second mathematical constraint to maximise fairness usually decreases accuracy. Because these constraints can be implemented in different ways, this trade-off is evaluated at the level of individual algorithms. It is also a concern for the causal approaches to fairness we will later discuss. Therefore, while it is an important problem for algorithmic fairness, it is not a shortcoming specific to the statistical approach definitions. This trade-off is discussed in papers presenting new methods and in comparative reviews (such as (44,89)).

The legitimacy of trading in some accuracy for fairness depends on our assumptions about what is just and about the relationships between the protected attribute, features, characteristics of interest, the target variable and decision. The incompatibility of several definitions likewise requires us to explicitly discuss these assumptions to justify the preference of one definition over others.

3.2.4 Worldviews

Data scientists have recently started reflecting on the causes for the lack of consensus (48–50,90). Friedler (48) first identified two main 'worldviews' underlying existing fairness definitions: 'We are all equal' (WAE) and 'What You See Is What You Get' (WYSIWYG).

According to WAE, innate differences between protected groups are rare, so (A) any difference in feature distribution is due to external factors, so (B) these features should not be used. WAE justifies definitions requiring equality in outcomes (e.g. Demographic Parity, Predictive Parity). WAE appears incomplete and makes very strong assumptions because it considers (A) and (B) as axioms, whereas (A) should be evidence-based – requiring causal relationships between features and protected attributes to be examined, and a normative moral principle should be used to justify going from (A) to (B). By not distinguishing the causes of discrimination, WAE confounds 'superficial' unfairness, where a group/individual's real features are misrepresented due to discrimination (e.g. a manager's racial prejudice), with 'structural' unfairness, where group membership progressively leads to a disparity in the real distribution of features .

To overcome the concern that accessible features sometimes reflect protected classes' characteristics of interest unequally (see standardised test-score example in Section 3.1, 'Feature Selection'), WYSIWYG assumes that accessible features reliably reflect characteristics of interest. WYSIWYG justifies definitions requiring equality of treatment (rather than outcome) (e.g. Fairness Through Awareness, Conditional Demographic Parity). Its assumption concerns causal relationships between features and characteristics of interest, but WYSIWYG ignores the existence or nature of causal relationships between protected attributes and features. Therefore, WYSIWYG would not correct for 'superficial' or 'structural' fairness.

These are rudimentary views because they do not justify their position or consider the nature of causal mechanisms between features and outcomes. Causal mechanisms are at the heart of discrimination and prediction. Discriminative processes unfairly change causal relationships, creating relationships (e.g. between 'race' and 'academic skills' via 'wealth'), or eliminating relationships (e.g. between 'relevant skills' and 'performance reviews' if managers are biased). Prediction-making relies on there being causal relationships between features and outcomes, because only then will features give reliable information about an outcome on the long term. ML predictions rely on correlations because they are much easier to extract from data, but data scientists use large and diverse datasets, hoping that extracted correlations reflect causal

relationships. Failing to consider causal relationships therefore leads to incomplete fairness definitions and flimsy justifications. We must make causal assumptions explicit before evoking normative views of justice to establish their fairness and finally debating the appropriate course of action. In Section 5 we will introduce the causal approach to fairness which shifts the focus to causal relationships between the protected attribute, characteristics of interest, target variable and outcome.

While the nature of discrimination and the limitations of statistical approaches suggest a causal approach, we consider whether normative approaches to Justice in philosophy and peoples' fairness perception also support this approach. Few statistical papers frame their view of fairness in the context of political philosophy or moral psychology; yet both are important when advancing a normative view of fairness.

4 FAIRNESS IN PHILOSOPHY AND PSYCHOLOGY

4.1 Philosophical approaches to Fairness

Questions at the heart of algorithmic fairness, including: "What is 'fair'?", "Is this definition context-dependent?", and "Which features can be fairly used despite disparate outcomes?" are also at the heart of political philosophy's theories of Justice. While statistical papers on algorithmic fairness echo these questions, they rarely discuss philosophical debates relevant to their definitions. Yet statistical approaches only provide the means to measure and implement certain views of fairness, not the justification and reasoning necessary to promote these normative views. Philosophical approaches therefore contribute a preliminary insight into the value and limits of different definitions or implementations.

We introduce four influential theories of Justice: Rawls' contractarianism, Scanlon's contractualism, Mill's utilitarianism, and Dworkin's luck egalitarianism, to give an overview of possible justifications (or lack thereof) for fairness-regarding measures and particularly for a causal approach to fairness. These theories are complex and regularly reinterpreted; we provide a simplified overview only.

4.1.1 Rawls' Contractarianism

The main principles of Rawls' Theory of Justice are the (91,92):

1. Liberty principle: respect of equal and extensive individual liberty, including freedom of conscience, association, expression, and democratic rights (e.g. freedom to vote and run for office)
2. Secondary principles:
    A. Equality of opportunity principle: People with the same capabilities and talent should have the same (economic and educational) opportunities regardless of their socio-economic status.
    B. Difference principle: inequalities are only permissible if they are advantageous to the least advantaged. Huge increases of inequality are therefore permissible if the least advantaged are better off than before.

The first principle takes precedence, so the second principles are only enforced if they do not encroach on the first (91,92). Rawls legitimises these normative principles by arguing that people would come to these same principles if they were to establish a social contract* to live by while behind a 'veil of ignorance', that is without knowing their position and characteristics in this society (e.g. race, gender, wealth, natural endowments etc.) (92). The theory's main criticisms concern the feasibility of the 'veil of ignorance' thought experiment (93), whether his second principles, if checked by the first, are sufficient to compensate for existing inequalities (e.g. disabilities) (38,94,95) and whether the vast inequalities permitted by the second principle are just (96).

Causal relationships matter in principle 2.A. to distinguish the effects of wealth and status on opportunities, but they are not a core element of this theory as both principles 1 and 2b can be evaluated with only minimal consideration for causality.

4.1.2 Scanlon's Contractualism

The moral principles that Scanlon advocates are those that no one could 'reasonably' reject as a the basis for common agreement (97). Individuals, without being behind a veil of ignorance, can veto a proposed principle if they consider that it would affect them unfairly. 'Justifiability' is the foundation of fairness: a principle is fair if it can be justified to others (92). 'Justifiability' is too flexible to endorse specific statistical definitions of fairness, but it does support trading in some accuracy for fairness (e.g. conditional statistical parity, excluding features which are not equally reliable across protected groups) since one can reasonably object to a measure which arbitrarily exempts some people from a burden carried by everyone else. By encouraging reflections on what constitutes legitimate grounds for the distribution of benefits (64), contractualism is relevant to identifying protected attributes and fair features. Finally, contractualism permits the effect of causal considerations (both causes and consequences) on individuals' sense of justifiability, but is not grounded in causal evaluation.

4.1.3 Mill's Utilitarianism

The above views were developed as alternatives to utilitarianism. The 'principle of utility*' is that we should act so as to produce the greatest good for the greatest number (98). However, this allows behaviour most people find immoral (e.g. arbitrary violence) if it brings about more utility for others than disutility for victims. Mill, an influential utilitarian, advocates in favour

of an adapted view of utilitarianism which prioritises rule consequentialism[5] (101): we ascribe to rules that have been designed by others who seek to maximise welfare for the greatest number and have considered short and long-term consequences (102). Beyond the consequences of an act, these rule-makers can consider how the permissibility of an act might influence society. This allows more accurate evaluations than if individuals had to judge consequences of an action independently and urgently (act consequentialism (103)).

Mill's utilitarianism depends on causal considerations: the consequences of a decision determine a its moral value. Indeed, the extent of causal reflections involved distinguishes rule and act consequentialism.

4.1.4 Egalitarianism

Different egalitarian views wish to establish equality based on different criteria, so discrimination (a form of inequality) is a central topic in egalitarianism and their reflections are directly relevant to quantitative approaches to fairness.

What kind of equality? And is this the same in all fields?

Are we interested in absolute equal distributions or in equal opportunities to compete for something? We may want an equal distribution of voting rights if people have an equal status as citizens (52). However, for social positions or economic goods, it might be fairer that individuals who are better qualified or put in more effort reap greater benefits, in which case we may want to ensure equal opportunity rather than an equal distribution. Therefore, the type of equality seems contingent on the circumstance.

Should we distinguish between luck and desert?*

This question is the foundation of a principle egalitarian theory: luck egalitarianism, developed by Dworkin (104). It distinguishes 'ambitions' and 'endowments'. 'Ambitions' relate to our choices (e.g. how much effort we put in or how we spend our money), 'endowments' to things beyond our control (e.g. genetic inheritance, family's socio-economic background,

[5] See (99,100) for different views on whether Mill is a rule consequentialist.

unforeseeable bad luck). According to luck egalitarianism, inequalities arising from different ambitions are permissible whereas inequalities arising from different endowments (e.g. bad luck) are not permissible and should be compensated (104). This echoes algorithmic fairness debates. For example, when debating whether it fair to use standardized-test scores greatly influenced by the amount of in-class practice (instead of subject knowledge), which depends on schools' wealth, we could untangle what corresponds to 'ambitions' and 'endowments' and apply the luck egalitarianism principles. However, distinguishing the role of ambitions and endowments is often difficult. If a child practices the piano diligently for years, it may partly be because they possess natural dispositions towards it (e.g. good pitch perception) and because their family could provide excellent classes.

Luck egalitarianism depends on causality: to understand the role of ambitions and endowments we must identify causal relationships between features. Dworkin's luck egalitarianism echoes the causal approach to fairness because it requires shifting the attention to causal relationships underlying the current situation, before applying normative principles of fairness.

4.2 Psychological approaches to fairness

Philosophical concerns provide justifications for normative stances and encourage useful questions, but implementing fairness also requires considering the fairness perception of individuals affected by algorithmic predictions (as emphasized by social contract* theories of justice). If algorithms are deemed fair only by data scientists unaffected by these algorithms, their fairness is questionable. Furthermore, perception of unfairness in algorithms can translate to mistrust of the institutions using them (e.g. a judicial system, a company).

We therefore present important findings regarding humans' perception of fairness. While informative, these findings pertain to human interactions and may not be generalisable to algorithms. We thus review two studies explicitly investigating algorithmic fairness perception.

4.2.1 Fairness perception in children

Studying fairness in children can reveal which fairness tendencies are innate and which depend on culture and complex reasoning. First, when young children are not personally involved they show pro-fairness tendencies, such as distributing resources equally (105), being surprised by unfair outcomes (106–108) and rewarding fair as opposed to unfair individuals (109,110).

Children also show pro-fairness tendencies when they are affected by the distribution but young children are more sensitive to disadvantageous inequality (when they are worse-off) then advantageous inequality (when inequality favours them) (111–114). Advantageous inequality aversion only appears in older children (115,116). A cross-cultural study (117) suggests that while disadvantageous inequality aversion is universal, the emergence of advantageous inequality aversion is culturally dependent. Young children (3-5 years old) also show merit-sensitivity when sharing resources: they give more to partners who has contributed more to a task, although their distribution remains tainted by disadvantageous inequality aversion as they always keep at least half the reward, even if they contribute less (118–120). However, these studies concern children in western societies and results cannot be generalized; in fact, one study (121) finds culturally-dependent merit-sensitivity in older children across Africa. Interestingly, (122) suggests that children's evaluation of merit in resource distribution scenarios depends on the causal relationships between individuals' contributions and the outcome of the game, which are modulated by changing the game's causal structure.

4.2.2 Fairness perception in adults

A canonical task to investigate fairness in adults is the ultimatum game in which a 'proposer' is given a sum of money which they must share however they like with the 'responder' who can then either accept the offer, or reject it, in which case neither gets any money. The classic ultimatum game is a one-off game between two participants who know nothing about each other. If players only seek to maximize their gain, a responder should accept any non-zero offer, and the proposer should therefore make the smallest non-zero offer possible. However, according to a meta-analysis (55), proposers offer 40% of the total sum on average (substantially more than the smallest non-zero offer), and responders reject 16% of offers[6]. The dictator's game has a similar setup though responders cannot reject offers. According to a meta-analysis (56), even without the threat of rejection, proposers offer 28.35% of the total sum on average. Behaviour in both games suggests adults care about fairness, even it means they incur losses. Responder behaviour is interpreted as rejecting a sum of money to enforce the social

[6] Proposers' offers are not culturally dependent, but responders' behaviour do change significantly across cultures although it cannot be accounted for by cultural traits (55).

norm of fairness (123) and proposer behaviour as reflecting pro-fairness tendencies, though tainted by a self-serving bias as they rarely split the offer equally.

Studies also find merit-sensitivity in adult fairness perception. When participant distribute resources to agents, there is a significant impact of these agent's decisions and bad luck on participant's choice of a fair distribution (57–59). In particular, participants are more likely to compensate an agent who encounters unforeseeable bad luck but less likely to compensate agents who make risky choices (e.g. in gambling) (58). Therefore, individual's agency in their outcome seems an important factor for human fairness perception. To assess agency, participants necessarily consider causal relationships between agents, external factors and the outcome. Indeed, modulating causal relationships between team members' contributions and the team's outcome (via the causal structure of the game) impacts how they assign responsibility to each other for the game's outcome (60). Though these studies only involve participants from Western countries and do not reflect universal characteristics of fairness perception, cultural fairness preferences are still useful in predicting whether certain algorithmic fairness methods will be considered fair in certain societies.

In conclusion, both children and adults care about fairness, albeit with a self-serving bias. Disadvantageous-inequity aversion appears innate while advantageous-inequity aversion appears more culturally dependent. There is also interesting evidence for the importance of causal relationships in fairness perception via children and adults' consideration for merit and agency.

4.2.3 Perceptions of algorithmic fairness

These conclusions support our interest in algorithmic fairness and in causality, but their results apply to human interactions, and may not generalize to decisions made by algorithms. We therefore briefly discuss the two papers which provide relevant insight to human perception of algorithmic fairness.

First, humans judge algorithms as unfair more frequently than a human making the same decisions, when these decisions have what might be considered an evaluative component (determining the value of a job candidate as opposed to work scheduling for example) (61).

The second paper (31) is directly relevant to our causal approach in Section 5 because it evaluates which features participants think can be fairly used to predict recidivism, and then

the reasons for which a participant considers these features fair. The more a participant found the feature in question 'relevant', 'reliable', 'volitional' and 'could cause the outcome', the more likely they were to find this feature fair. Conversely, if they found that using the feature 'could cause a vicious circle', then they were more likely to consider it unfair. Participants had a similar way of using these properties when determining the fairness of a feature. However, they then attributed properties to features very differently, which explains their different opinions of features' fairness. They particularly disagreed on which features 'could cause the outcome'. This suggests that causality is a crucial aspect for people's fairness evaluations (so much so that it accounts for part of their disagreements) but that people rarely agree on the nature of causal relationships. Therefore, it is important to consider causality, but relying on specific causal relationships to justify the use or non-use of certain features (as we do in Section 5) can prove controversial. However, this study has important limitations: it is the only one on this subject, it only considers fairness judgements in one context (recidivism prediction), and distinguishing 'relevance' and 'reliability' from causal considerations is disputable – some may rely on causal relationships to judge a feature's relevance and reliability.

In conclusion, there is no philosophical consensus which could justify one statistical definition of fairness, but several philosophical views place importance on causal relationships. Humans seem to display an innate and ego-centric fairness preference, which, in Western cultures at least, also seems agency and merit-sensitive, and thus requires a consideration of relevant causal relationships.

5 A CAUSAL APPROACH TO ALGORITHMIC FAIRNESS: COUNTERFACTUAL FAIRNESS

In Section 3.2 we show that the lack of consensus regarding statistical definitions stems from their creator's different implicit assumptions concerning both the nature of the relevant causal relationships, and what constitutes a fair causal relationship. A common ground regarding underlying causal relationships must be established before debating their fairness, identifying the fair course of action, and finally choosing an implementation method. Because assumptions are implicit, it is difficult to understand why definitions fail in some circumstances, and to provide targeted criticism. Definitions are therefore rejected when faced with a problematic example.

Causal approaches to fairness focus on ensuring that algorithms do not rely on data which reflects unfair causal relationships. They also distinguish between the steps mentioned above. This greater transparency allows targeted criticism and greater flexibility. Indeed, causal approaches can be adapted to different circumstances and updated in light of new causal knowledge and new fairness preferences.

We first introduce Pearl's recently developed framework for formalising causal relationships on which relies the causal approach to algorithmic fairness, before presenting an application: Counterfactual Fairness (CF), which reveals some of the causal approach's advantages and limitations.

5.1 Causal inference in statistics

5.1.1 Introduction to causal inference in statistics

In 2000, Judea Pearl presented a mathematical foundation to express and analyse causal relationships, driven by statistics' inability to answer pressing questions such as "Does increasing police presence in this neighbourhood prevent crime?" or "Is this employer guilty of including sexual discrimination in their hiring policy?", which concern cause-effect relationships (66). Indeed, non-causal data and equations cannot distinguish the cause and the effect. For example, if a thermometer's mercury rises when the weather is warm, statistics

alone cannot tell whether the increased mercury levels causes warm weather or vice versa. Pearl's 'Structural Causal Models' allow us to represent causal relationships through both causal diagrams and symbolic language (like algebra) (66). Causal diagrams (c.f. *Figure 3*) are composed of nodes and edges. Nodes represent a feature, such as 'A-level grade' or 'gender'. Edges are directional and represent a process that mediates a causal relationship between the issuing and receiving nodes. In *Figure 3*, nodes A and B are the 'parents' of node C and node C is the 'descendent' of nodes A and B.

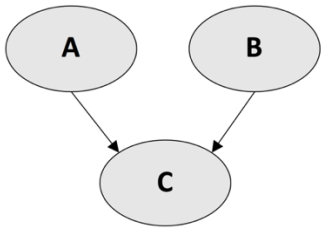

Figure 3: A simple causal model

Since Pearl's 2000 book (124), a range of mathematical methods have been developed to infer causal relationships from experimental* and observational* data and to test claims of causal relationships (see (64) for an overview).

Causal models also allow us to use reasoning methods that we cannot apply to correlational data. 'Explaining away' is an example of a probabilistic reasoning method made possible by causal graphs. In *Figure 3*, though nodes A and B are independent, in some circumstances, knowing the causal relationships involved and the values of nodes A and C can give us information about node B in some circumstances (125,126). For example, if A and B are two possible causes of C, and we know that C occurred and A did not occur, we increase the probability that B occurred. In Section 5.2.2, Example 1, we show how 'explaining away' allows us to obtain fairer data.

5.1.2 Ladder of Causation

Pearl classifies three levels of cognitive ability: 1. seeing, 2. doing and 3. Imagining linked to three levels of understanding causal relationships: the three rungs on his 'Ladder of Causation' (66) :

1. **Association**: Two events are associated if the likelihood of observing one event is changed when we have observing the other (66). For example, does being of a certain race inform the likelihood of being hired in this company?

2. **Intervention**: Intervention requires deliberately changing a specific state of things and observing the effect on something else. Suppose we intervene to send a cohort of African-American students (representative of the various income-levels within the USA's African-American community) to schools in wealthy neighbourhoods; will their average SAT grades be higher than African-American students' current average SAT grades?. Interventions are a common tool for making causal inferences; drug trials, for example, use interventions to infer the efficacy of drugs.

3. **Counterfactuals**: Beyond seeking an event's causes and effects, we might ask counterfactual questions such as "Had the past been different, would the present be different too? And if so, how?". Counterfactuals allow us to ask and answer questions that are incompatible with reality such as "Would the current average income of African Americans be higher if school funding in the USA did not depend on local wealth?". We can make counterfactual assertions by virtue of causal relationships' generalizability. If a causal relationship is established between education quality and future wealth, and we know African Americans attend poorer and so worse schools than white Americans on average, then we can generalize this relationship to answer the above counterfactual question. To make counterfactual assertions in complex, real-world situations, one must build a causal model, then perform an intervention (for example removing the relationship between local wealth and school quality), then allow this change to propagate down the causal model in order to finally observe its effect on the outcome. Counterfactuals are at the top of the ladder because they require the two previous rungs, and because of their immense value: counterfactual thinking impacts our capacity to "reflect and improve on past actions and, perhaps even more significant, our willingness to take responsibility for past and current actions." (66), two functions which are crucial for ensuring algorithmic fairness.

A causal model, presenting the causal relationships relevant to a particular situation, bridges the gap between passive observations, predicting the effects of interventions, and identifying what could have been.

Causal approaches to fairness focus on ensuring that algorithms do not rely on data which reflects unfair causal relationships. However, they do not provide a normative approach for judging whether causal pathways are fair. For this we must rely on philosophical reflections and theories of justice. The causal approach makes us question the fairness of causal

relationships underlying the data, and provide methods to implement fairness once we have identified fair and unfair pathways and features.

5.2 Counterfactual Fairness

5.2.1 Definition

To illustrate the causal approach to algorithmic fairness, we examine the first and most discussed causal definition of algorithmic fairness: Counterfactual Fairness (CF), introduced by the Alan Turing Institute's 'Counterfactual Fairness' group in 2017 (32). CF requires making the assumptions about causal relationships within the data explicit via a causal model, and provides an intuitive definition and implementation of fairness. We define CF and use examples to understand its implications. We then consider why its advantages over non-causal statistical definitions of fairness and finally discuss the its limitations and possible causal alternatives.

An algorithm's prediction for an individual is counterfactually fair if it would have been the same in a counterfactual world where the individual's protected attribute was different. The definition is intuitive because it echoes questions that individuals ask when questioning the fairness of a situation: "would I have been treated differently had I been of a different gender?", "would the outcome be the same had I been of a different race?". It is an individual-level definition which echoes people's ego-centric perception of fairness discussed in Section 4.2.

CF can be expressed mathematically thus:

$$P\big(\hat{Y}(a) = y \mid X = x, A = a\big) = P\big(\hat{Y}(a') = y \mid X = x, A = a\big)$$

Equation 1: equation for CF(drawn from (68))

Note: Upper case letters represent variables, lower case letters represent instances of these variables.

$\hat{Y}$ is the algorithm's prediction of the probability of a certain target variable. A is the protected attribute of an individual, which is chosen by the people creating or requesting the algorithm. Race and gender, which we use in examples, need not be protected attributes when there is a legitimate reason for including them. X represents the other observable attributes of each individual.

Equation 1 compares the probability of an individual with features a (the protected attribute) and x of receiving the prediction y, to the probability of the same individual receiving the same prediction y if their protected attribute had been different (a'). Changing a to a' is the intervention performed on the model but we obtain the counterfactual ($\hat{Y}(a')$) by allowing this change to propagate down the model to the target variable (if there is a causal pathway between the protected and target variable).

A prediction is therefore counterfactually fair if the probability of an individual receiving this prediction would have been the same in a similar world where we only changed their protected attribute, but allowed this change to propagate and impact downstream features, according to their causal relations.

5.2.2 Illustrating Counterfactual Fairness

To illustrate the implications of CF, we consider examples of causal diagrams of features provided to algorithms whose predictions inform decisions. The causal models used here simplify complex real-world scenarios and we do not claim their validity.

Example 1: Using a test score to predict passing first year exams at university

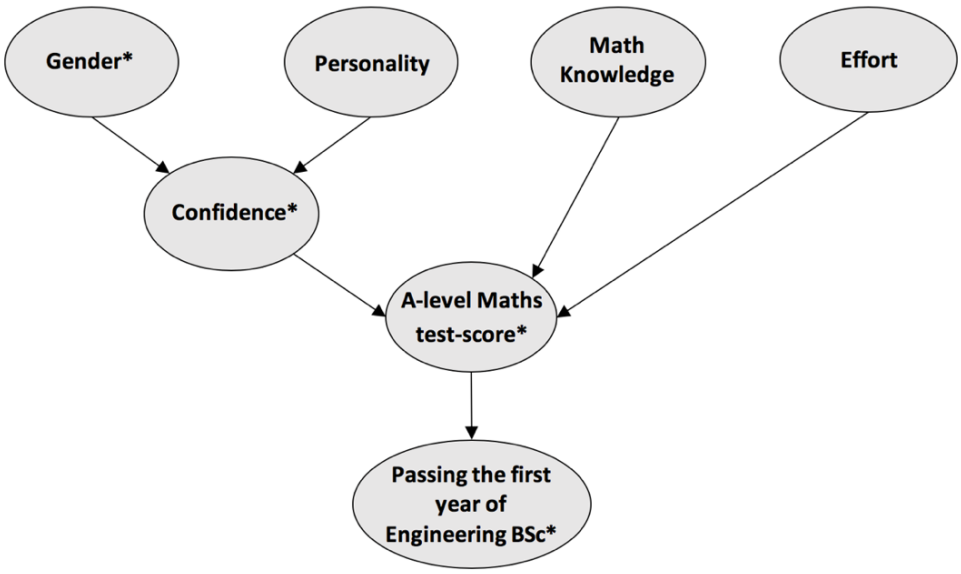

Figure 4: Causal model for the prediction of an applicant's likelihood of successfully completing a Bachelor (BSc) of Engineering

Note: (*) identifies variables that can change in the counterfactual experiment.

An algorithm is given ‘A-level maths test score’ (among other features) to determine admission to a university engineering course. The admissions team is interested in the candidate's knowledge of mathematics, and this test-score seems to be an accessible proxy for them. However, when plotting the causal graph, we might find that ‘confidence’ impacts test performance, and that ‘confidence’ is closely linked to wider group representation. In this graph, different levels of ‘confidence’ mediate an effect of gender on the test score. If Student 1 with gender A was rejected from the course, they might ask : “would I had been accepted if I had been of gender B?” This counterfactual involves changing Student 1’s ‘gender’ feature and propagating this change’s effects on the variables causally downstream, including ‘confidence’ and ultimately ‘A-Level maths test-score’, but not ‘math knowledge’, which is independent of ‘gender’ (according to the model). If the admissions team considers gender to be a protected attribute, and calculating the counterfactual shows that Student 1 had a 0.4 probability of being accepted in the real world, but a 0.6 probability of being accepted in the counterfactual world, then the prediction is not counterfactually fair.

Even without implementing CF, the causal model and ‘Explaining Away’ could be used to improve fairness. Supposing the admissions team is interested in the test-score only because it reflects ‘math knowledge’ and ‘effort’, and does not care about ‘confidence’. Lower confidence somewhat confounds test-scores’ representativeness of ‘math knowledge’ and ‘effort’ for students of gender A. ‘Explaining away’ tells us that a test-score of S reflects higher ‘math knowledge’ and ‘effort’ for a student with gender A, than for a student with gender B. So ‘explaining away’ allows us to correct our understanding of students’ features of interest.

Example 2: Predicting the risk of recidivism

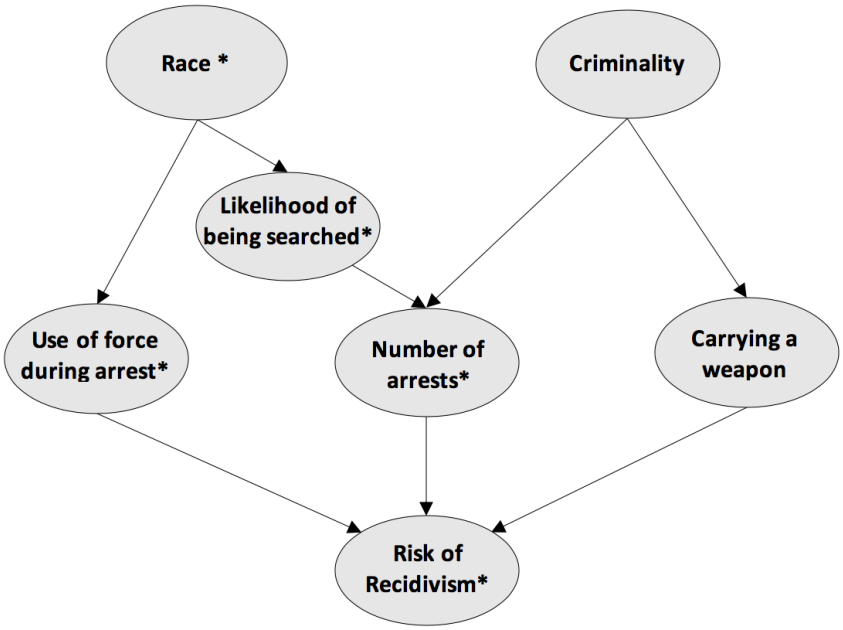

Figure 5: Causal model for the prediction of recidivism

Note: 'Criminality' is an abstract feature (characteristic of interest) that indexes race-independent characteristics used by the police. (*) identifies variables that can change in the counterfactual experiment.

An algorithm uses the features: 'number of arrests', 'use of force during arrest' and 'carrying weapon' (at time of arrest) to calculate recidivism likelihood if given bail. However, if the causal model above is valid and race is considered a protected attribute, then using 'number of arrests' and 'use of force during arrests would mediates a counterfactually unfair effect of race onto the risk of recidivism.

If Prisoner 1 of race A was denied bail due to the algorithm's estimation of high recidivism risk, they might ask: "Would I have received the same estimation if had I been of race B?". A counterfactual analysis would change 'race' in the diagram, and propagate this change by altering 'use of force during arrest', 'likelihood of being searched by a police officer' and 'number of arrests', but not 'criminality' or 'carrying a weapon'. If Prisoner 1 of race A has a 0.8 probability of being found to have a high risk of recidivism, and a 0.8 probability in the counterfactual scenario when they are of race B, then the algorithm's outcome is not counterfactually fair.

5.3 Critical assessment of Counterfactual Fairness

5.3.1 Advantages

Providing an implementation method

While previous statistical definitions only provide a measure of fairness and can sometimes be satisfied through unfair processes (Demographic Parity for instance), CF provides a method for implementing itself. Equation 1 implies that the predictor* $\hat{Y}$ is counterfactually fair if it is not dependent on any descendants of the protected attribute (32). Removing features that are the protected attribute's descendants is the most straightforward way to satisfy CF. In Example 2, according to the model, 'number of arrests' and 'likelihood of being searched' are descendants of 'race' so using these features would lead to counterfactual unfairness, but 'criminality' is causally independent of race so an algorithm that relied only on 'criminality' or its descendent 'carrying a weapon' would be counterfactually fair.

This sometimes removes many observable features, a common method is to rely on relevant latent variables instead (like 'math knowledge' and 'criminality'). Latent variable are used to represent non-observable properties that are difficult to measure directly. They are a common tool used across disciplines (psychology, economics, social sciences, AI etc.) and many mathematical methods have been developed to infer them from observable variables.

In addition, because causal models can be used to predict the effects of interventions (68), the same causal models built to inform feature selection for an algorithm can be used to predict the effects of different fairness implementations (126), for example to evaluate the benefits of positive discrimination.

Making assumptions explicit

CF requires the specification of a causal model which makes explicit the assumptions about causal relationships underlying the data. By first requiring decision-makers to agree on the causal links between the true characteristics of interest, the features used, and the predicted property, it creates a valuable common ground for subsequent interrogations about the fairness of features and fairness of the pathways between a potential protected feature and the predicted property.

Comparison to statistical definitions of fairness

Interestingly, CF often subsumes some statistical definitions of fairness. For example, if there are no causal paths between the predictor and the protected attribute, then we have $\hat{Y} \perp A$, therefore Demographic Parity ($\hat{Y} \perp A$), Equalised Odds $\left(\hat{Y} \perp A \middle| Y\right)$ and Calibration $\left(Y \perp A \middle| \hat{Y}\right)$ are satisfied. Equalized Odds and Calibration could be satisfied even when the predictor and protected attribute are dependent, though not simultaneously. But according to the causal approach to fairness, satisfying Equalized Odds or Calibration in this circumstance would mean permitting unfair causal relationships between the protected attribute and the prediction. Therefore, in the CF framework, these two definitions' incompatibility in this circumstance is irrelevant since neither would guarantee fairness (37,126).

CF also improves upon Fairness Through Awareness. Both are individual-levels definitions, but because the latter evaluates fairness by comparing individuals with similar features, it only considers unfair a disparate use of features to predict an outcome, thus excluding the possibility of unfair relationships between the features and protected attribute. CF overcomes this problem by allowing the change in protected attribute to propagate down the model through causal relationships. CF can therefore compensate for a wider range of unfair situations than Fairness Through Awareness, while maintaining the advantage of an individual-level definition which fits people's ego-centric fairness perception.

Finally, CF improves on Fairness Through Unawareness. In the latter, algorithms must not use the protected attribute to make predictions, but the protected attribute is often reflected in other features, making this definition insufficient. CF takes this into account by removing features causally downstream from (so possibly correlated with) the protected attribute.

5.3.2 Limitations of the Counterfactual and Causal Approach

Causal models require untestable assumptions

"All models are wrong but some are useful" – George Box

The principal criticism CF faces concerns the reliability of the causal models it requires. Causal models integrate data and expert knowledge to understand the complex mechanisms underlying a final outcome. They can be increasingly plausible as data is found to coincide with their predictions rather than those of alternative models, but, like all models, cannot be proven

correct. While experimental* studies are the gold standard for drawing causal conclusions, not all interventions are possible or ethical, so most causal models rely on observational* data. Observational data can mask confounding factors*, but several methods of 'sensitivity analysis' have been developed to estimate the amount of unmeasured confounding in causal models (see (127–129) for further information). Bridging the gap between observational data and causal models also requires making assumptions, including some untestable ones (67,69,130). Therefore, experts can disagree on the right causal model and decision-makers can dispute their validity, which complicates CF's implementation. However, this very process, by requiring the parties involved to look for evidence to justify their beliefs about a situation's underlying causes, is also the principle benefit of CF because it forces the parties to make more informed decisions.

Nevertheless, to overcome the problem of having multiple plausible causal models, the proponents of CF put forward a new definition: Approximate Counterfactual Fairness (ACF) (67), which tolerates a small gap between an individual's prediction in the real and counterfactual world. This slightly relaxed fairness definition allows for predictions that are considered fair with respect to several causal models at once. This method has the additional advantage of providing a better control over the fairness-accuracy trade-off by adding the fairness constraint to the algorithm's loss function* so that during the learning process, the algorithm simultaneously minimizes prediction errors and unfairness (maximizes accuracy and fairness).

While causal models have limitations, they also expose assumptions underlying algorithms' predictions, leaving these (assumptions and subsequent implementations of fairness) open to criticism and revision. Beyond increasing transparency, this enables the field of algorithmic fairness to move past its current deadlock regarding statistical fairness definitions, which (unlike CF and ACF) rely on implicit assumptions and must be disregarded rather than updated if underlying assumptions do not hold.

Counterfactual Fairness is not flexible enough

CF is also criticized for its systematic removal of all features causally downstream of a protected attribute. Data scientists and decision-makers might consider certain causal pathways between the protected attribute and outcome as fair and wish to use features in these pathways to inform the prediction process. For example, if confidence is a crucial part of a job (e.g. public

speakers), a hiring panel may find it fair to include a feature that reflects 'confidence', even if this feature is causally downstream from 'gender' (c.f. *Figure 4*), even while other features causally downstream from 'gender' are still deemed 'unfair'. CF considers all pathways between the protected attribute and outcome as unfair, but Path-specific Counterfactual Fairness (36), another causal approach to algorithmic fairness, enables the distinction of fair and unfair pathways between the protected attribute and outcome. According to Path-specific CF, a decision is unfair if it would be different in a counterfactual world where the change of protected attribute has been propagated only along aforementioned unfair pathways. Chiappa & Gillam (36) ensure Path-Specific CF by correcting features in unfair causal pathways rather than removing them altogether.

6 Conclusion

The predictions of algorithms are increasingly used to inform important decisions, but they face the challenge of inherently biased data. In the past three years, publications presenting definitions and implementation methods to ensure algorithmic fairness have grown exponentially. However, the field has recently found itself unable to agree on a definition of fairness, which is problematic because many definitions are incompatible. This incompatibility arise because data scientists base their definitions on beliefs about underlying causal relationships and on assumptions about what is 'fair' which are not made explicit. Therefore, they do not share a common understanding of the world on which to ground a debate on the legitimacy of different fairness implementations.

We argue that the recently developed causal approach is more appropriate than conventional statistical approaches which cannot capture causal relationships. Indeed, causal relationships are central elements of the processes of discrimination and prediction (as seen in Section 3.2.3). The approach is also supported by humans' consideration for merit and agency when making 'fair' distributions and by several philosophical theories of justice, which consider the consequences and/or causes of an action or situation as crucial elements for evaluating its fairness.

Furthermore, the causal approach, rather than adding to the mass of controversial definitions, allows the field to move past its current stalemate. It shifts the focus to the causal relationships underlying the data: it requires data scientists/decision-makers to make them explicit, then to evaluate their fairness, before debating the fair course of action and applying a chosen method for dealing with unfair relationships (e.g. removing or correcting the features affected by the latter). This increases the transparency and flexibility of the process, which facilitates targeted criticism and makes it amenable to change (e.g. updating the causal model, changing implementation method etc). Because previous statistical approaches do not distinguish these stages, the reason for their respective shortcomings is harder to identify; therefore, if they do not guarantee fairness in some circumstances (as seen in 3.2.1-3), they cannot be amended or updated, and must be discarded.

The causal approach does not resolve the nature of fairness; it provides the tools to apply different views of fairness to different situations, whereas statistical definitions provide the

tools to implement specific views of fairness in specific circumstances (both implicit). Defining fairness requires complex considerations, such as finding a legitimate basis for social justice and considering all the contexts relevant to fairness (e.g. distributive justice, criminal justice). Therefore, beyond a causal approach, the field of algorithmic fairness requires closer interactions between data scientists, philosophers and sociologists (to establish causal links).

7 Appendix

7.1 Glossary

Agency: an individual's capacity to act, and choice of exercising this capacity.

Algorithms: a finite set of mathematical steps often given to a computer to complete a specific task.

Anchoring effect: "the anchoring effect is a cognitive bias that describes the common human tendency to rely too heavily on the first piece of information offered (the "anchor") when making decisions. During decision-making, anchoring occurs when individuals use an initial piece of information to make subsequent judgments. Once an anchor is set, other judgments are made by adjusting away from that anchor, and there is a bias toward interpreting other information around the anchor." (131)

Availability heuristic: a heuristic whereby people make judgments about the likelihood of an event based on how easily an example, instance, or case comes to mind. (132) For example, doctors may judge the efficacy of a drug or the risks of complications based on their most recent experiences with patients (133).

Base rates: the probability of a member in a particular group to show a certain outcome, irrespective of their features. E.g. the probability of female students of being accepted in a certain university degree.

Characteristics of interests: features a decision-maker is interested in but that are difficult to measure and so evaluated indirectly through more accessible features; for example 'Math Knowledge' can be evaluated through 'Math A-level score'.

Confirmation bias: a cognitive bias whereby people form a judgement and then look for and remember only the information that supports this initial judgement (132).

Confounding factor (or factor): a variable that influences both the independent variable and the dependent variable. An independent variable can seem closely correlated to a dependent variable, raising questions about their causal relationships, when in fact the association is mediated by a confounding variable.

Desert: what someone deserves in virtue of something else (134).

Experimental data Vs. Observational data: in observational studies, researchers gather data on existing circumstances, without interfering. This means that they cannot isolate dependent and independent variables, so there are many possible confounding factors interfering with the factor they are studying. Experimental data is drawn from experimental studies where researchers interfere strongly, often by controlling a number of external factors, which become

independent variables, to understand their effects on the dependent variables. In experimental studies, because experimenters can interfere with one independent variable at a time, it is easier to make causal inferences from experimental data than from observational data, although the possibility of unknown confounders still remains.

Features: a measurable property of an examples within the data, e.g. a prisoner's number of arrests, a student's SAT score, a job applicant's highest level of education etc.

Framing effect: heuristic whereby people's choices are influenced by the way the options are framed, through different wordings and setting, e.g. doctors tend to interpret mortality rates differently when they are framed in terms of survival percentage or mortality percentage (135).

Halo effect: "phenomenon whereby we assume that because people are good at doing A they will be good at doing B, C and D (or the reverse—because they are bad at doing A they will be bad at doing B, C and D). The phrase was first coined by Edward Thorndike, a psychologist who used it in a study published in 1920 to describe the way that commanding officers rated their soldiers. He found that officers usually judged their men as being either good right across the board or bad. There was little mixing of traits; few people were said to be good in one respect but bad in another." (136)

Loss function: quantify the algorithm's error (aka the 'loss') by comparing its outputs to their target values, usually obtained from past data (137). Many ML methods require updating the algorithm's model such that the loss function is minimized.

Machine learning (ML): the study of statistics and computer algorithms aiming to allow algorithms to automatically learn from data and improve their performance without being explicitly programmed (138).

Outcome: the reality behind the target variable. When training, algorithms generally compare their prediction of the target variable to the actual outcome found in the training data.

Positive outcome/prediction: the outcome/prediction that results in a positive answer to the question the algorithm's prediction must answer. For example, if the algorithm must evaluate whether a prisoner <u>will reoffend</u>, then the positive outcome/prediction is that the prisoner <u>did</u>/<u>will reoffend</u>. If the algorithm must evaluate whether a prospective student <u>will pass</u> their first year exams, then the positive outcome/prediction is that the student <u>did/will pass</u> their exams.

Principle of utility: the idea that we can judge how 'good' an action is, or which action out of two is the best, by considering how much value (utility) these actions bring to people – both in terms or how much utility they bring to individuals and how many individuals they affect. The definition of utility is itself very controversial. It can sometimes be understood as 'happiness' but can also encompass more complicated ideas such as long-term benefits in spite of short-term discomfort.

Protected feature (or attribute): a feature whose influence on the outcome is considered unfair; race and gender are sometimes considered protected attributes. A '**protected class (or group)**" is any group defined by the possession of a protected attribute. If 'race' is the protected attribute, protected classes could be binary E.g. 'White' and 'Non-white', or plural 'White', 'Asian', 'Black', 'Hispanic' etc.

Race: We use the word 'race' in terms of the widespread perception of different races which affects the treatment of groups of people thus characterised.

The existence of genetic foundations for 'races', as identified by the public, and the evidence for different cognitive capacities between these 'races' are strongly debated topics in academic circles. These topics would warrant an in-depth investigation, involving a thorough literature review and critical analysis of statistical methods used, which are both beyond the scope of this dissertation.

Considering that there is no clear scientific consensus about the legitimacy of the concept of 'race' or about the stereotypes associated to different 'races' (e.g. about intelligence and character traits), we cannot and do not accept either notion. Instead, we focus on the indubitable effect that the *perception* of race (relying mainly on visual cues, such as skin colour, and cultural characteristics, such as names) can have on the treatment of people, both on the short term (e.g. immediate discrimination during an interview) and on the long term (e.g. pervasive discrimination, throughout the job industry, during children's education, when granting loans/mortgages etc.) (123,139,139–146). These stereotypes cause certain 'races' to be poorer on average, in turn creating a host of difficulties (e.g. less access to education, less opportunities for personal development, more responsibilities, more difficult family lives etc) which create a 'hereditary' disadvantage correlated with 'race'.

Social contract: relies on the idea that "the agreement of all individuals subject to collectively enforced social arrangements shows that those arrangements have some normative property (they are legitimate, just, obligating, etc.)" (147). The resulting set of social arrangements constitutes this 'social contract'.

Supervised Learning: "the process of teaching a model by feeding it input data as well as correct output data. This input/output pair is usually referred to as 'labelled data'. Think of a teacher who, knowing the correct answer, will either reward marks to or take marks from a student based on the correctness of her response to a question."(148) This contrasts with Unsupervised Learning where the data does not contain the correct output for a given input. Supervised and Unsupervised learning are generally used for different kinds of problems.

Target variable: what the algorithm tries to predict in order to inform a certain decision. E.g. 'Likelihood of recidivism' and 'likelihood of passing first year exams' can be target variables used respectively to inform decisions about granting bail and college admissions.

Training data: The algorithm is given a set of data to allow it to create an inner model of how the features can be used to predict the output. In the case of supervised learning the training

data contains both an input (an example and its features) and its corresponding correct output. For example, an algorithm can be given examples of past prisoners and their features (e.g. the nature of their crime, their current sentence and their criminal record); it tries to use the features to predict whether these prisoners would reoffend if given bail and then compares its predictions to prisoners past outcomes (did or did not reoffend) to learn from its mistakes.

Weights: the importance given to each feature in the algorithm's inner model of how useful features are in predicting an outcome.

7.2 Statistical measures of fairness

7.2.1 Accuracy metrics

Cited from Verma & Rubin's "Fairness Definitions Explained" (47):

	Actual – Positive	Actual – Negative
Predicted – Positive	**True Positive (TP)** PPV = $\frac{TP}{TP+FP}$ TPR = $\frac{TP}{TP+FN}$	**False Positive (FP)** FDR = $\frac{FP}{TP+FP}$ FPR = $\frac{FP}{FP+TN}$
Predicted – Negative	**False Negative (FN)** FOR = $\frac{FN}{TN+FN}$ FNR = $\frac{FN}{TP+FN}$	**True Negative (TN)** NPV = $\frac{TN}{TN+FN}$ TNR = $\frac{TN}{TN+FP}$

"1. True positive (TP): a case when the predicted and actual out- comes are both in the positive class.

2. False positive (FP): a case predicted to be in the positive class when the actual outcome belongs to the negative class.

3. False negative (FN): a case predicted to be in the negative class when the actual outcome belongs to the positive class.

4. True negative (TN): a case when the predicted and actual outcomes are both in the negative class.

5. Positive predictive value (PPV): the fraction of positive cases correctly predicted to be in the positive class out of all predicted positive cases, $\frac{TP}{TP+FP}$. PPV is often referred to as precision, and represents the probability of a subject with a positive predictive value to truly belong to the positive class, $P(Y = 1 \mid d = 1)$. In our example, it is the probability of an applicant with a good predicted credit score to actually have a good credit score.

6. False discovery rate (FDR): the fraction of negative cases incorrectly predicted to be in the positive class out of all predicted positive cases, $\frac{FP}{TP+FP}$. FDR represents the

probability of false acceptance, $P(Y = 0 \mid d = 1)$, e.g., the probability of an applicant with a good predicted credit score to actually have a bad credit score.

7. False omission rate (FOR): the fraction of positive cases incorrectly predicted to be in the negative class out of all predicted negative cases, $\frac{FN}{TN+FN}$. FOR represents the probability of a positive case to be incorrectly rejected, $(P(Y = 1 \mid d = 0))$, e.g, the probability of an applicant with a bad predicted credit score to actually have a good score.

8. Negative predictive value (NPV): the fraction of negative cases

correctly predicted to be in the negative class out of all predicted

negative cases, $\frac{TN}{TN+FN}$. NPV represents the probability of a subject with a negative prediction to truly belong to the negative class, $P(Y = 0 \mid d = 0)$, e.g., the probability of an applicant with a bad predicted credit score to actually have such score.

9. True positive rate (TPR): the fraction of positive cases correctly predicted to be in the positive class out of all actual positive cases, $\frac{TP}{TP+FN}$. TPR is often referred to as sensitivity or recall; it represents the probability of the truly positive subject to be identified as such, $P(d = 1 \mid Y = 1)$,. In our example, it is the probability of an applicant with a good credit score to be correctly assigned with such score.

10. False positive rate (FPR): the fraction of negative cases incorrectly predicted to be in the positive class out of all actual negative cases, $\frac{FP}{FP+TN}$. FPR represents the probability of false alarms –falsely accepting a negative case, $P(d = 1 \mid Y = 0)$, e.g., the probability of an applicant with a actual bad credit score to be incorrectly assigned with a good credit score.

11. False negative rate (FNR): the fraction of positive cases incorrectly predicted to be in the negative class out of all actual positive cases, $\frac{FN}{TP+FN}$. FNR represents the probability of a negative result given an actually positive subject, $P(d = 0 \mid Y = 1)$, e.g., the probability of an applicant with a good credit score to be incorrectly assigned with a bad credit score.

12. True negative rate (TNR): the fraction of negative cases correctly predicted to be in the negative class out of all actual negative cases, $\frac{TN}{FP+TN}$. TNR represents the probability of a subject from the negative class to be assigned to the negative class, $P(d = 0 \mid Y = 0)$, e.g., the probability of an applicant with a bad credit score to be correctly assigned with such score."

7.2.2 Mathematical equations of statistical definitions of fairness

In complement of Section 2.2.

Notations:

$\hat{Y}$: predicted probability for an individual to belong to a certain label e.g. predicted probability for a prisoner to be 'very likely' to reoffend

Y: actual probability of a certain classification e.g. real probability for a certain prisoner to be 'very likely' to reoffend

$\hat{C}$: predicted classification for an individual, usually assigned when $\hat{Y}$ is above/below a certain threshold

A: protected attribute, here considered binary $A \in \{0,1\}$ e.g. male or female

X: all none-protected attributes made available to the algorithm

Z: a subset of X considered as features that can legitimately affect the prediction $\hat{Y}$

x_i: a given individual

D: a metric measuring the difference between predictions

d: a metric measuring the difference between two individuals (based on their features)

Demographic Parity: A predictor $\hat{Y}$ satisfies Demographic Parity if

$$P(\hat{Y} = y \mid A = 0) = P(\hat{Y} = y \mid A = 1)$$

Conditional Statistical Parity: A predictor $\hat{Y}$ satisfies Conditional Statistical Parity if

$$P(\hat{C} = 1 \mid Z = z, A = 0) = P(\hat{C} = 1 \mid Z = z, A = 1)$$

Predictive Parity: A predictor $\hat{Y}$ satisfies Predictive Parity if

$$P(Y = 1 \mid \hat{C} = 1, A = 0,) = P(Y = 1 \mid \hat{C} = 1, A = 1)$$

Equal Opportunity (or Equality of Opportunity): A predictor $\hat{Y}$ satisfies Equal Opportunity if

$$P(\hat{Y} = 1 \mid Y = 1, A = 0) = P(\hat{Y} = 1 \mid Y = 1, A = 1)$$

Equalised Odds: A predictor $\hat{Y}$ satisfies Equalised Odds if

$$P(\hat{Y} = 1 \mid Y = y, A = 0) = P(\hat{Y} = 1 \mid Y = y, A = 1) \text{ where } y \in \{0,1\}$$

Calibration: A predictor $\hat{Y}$ satisfies Calibration if

$$P(Y = 1 \mid \hat{Y} = y, A = 0) = P(Y = 1 \mid \hat{Y} = y, A = 1)$$

Fairness Through Awareness (or Individual Fairness): A predictor $\hat{Y}$ satisfies Fairness Through Awareness if

$$D(\hat{Y}(x_a), \hat{Y}(x_b)) \leq d(x_a, x_b) \quad \forall\, x_a, x_b \quad (149)$$

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